JIEMING ZHANG

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EDUCATION

University of Virginia **Charlottesville, VA**

B.S. Computer Engineering & B.S. Electrical Engineering, ***GPA: 3.9*** *Expected May 2029*

Relevant Coursework: Data Structures & C++, Computer Systems Organization I, CPU Architecture, Digital Logic Design

TECHNICAL SKILLS

Languages: Python, C++, Java, VHDL, Assembly

Software & Web: Playwright, Beautiful Soup, Chromium, HTML, Local Web Servers, Git

Data & Visualization: Matplotlib, SciPy, CSV Processing, Data Cleaning

Systems: Linux, Vim, ESP32, I2C, SPI

TECHNICAL PROJECTS

Assignment Dashboard **Spring 2026**

Python Automation & Local Web Application | Personal Project

- Built a local assignment dashboard that automatically scraped Gradescope and Canvas with Playwright/Chromium and Beautiful Soup, consolidating coursework in one interface launched at login through Windows Task Scheduler.
- Parsed and normalized assignment deadlines with *dateutil*, removed duplicate records, surfaced date-based warnings, and enabled sorting by course or due date.
- Added one-click links into each source assignment and a user-triggered rescrape workflow; stored browser session cookies outside version control through *.gitignore*.

IoT Golf Swing Analyzer & Visualizer Prototype **Summer 2026**

Python Analytics & Embedded Data Pipeline | Personal Project

- Created a Python telemetry engine using Matplotlib and SciPy to process raw IMU quaternions, correct gravitational drift, estimate clubhead velocity, and render 3D swing trajectories.
- Designed C++ firmware and an ESP32 data pipeline that recorded 100 Hz IMU data into CSV files for offline analysis.

Toy ISA Instruction Set & Assembly Programming **Fall 2026**

Computer Systems Organization I | UVA

- Translated hand-encoded hexadecimal programs into assembly for a toy 8-bit ISA, working with registers, stack operations, and conditional control flow on a Linux environment.

ENGINEERING EXPERIENCE

Virginia Motorsports (Formula SAE) **Fall 2025 - Present**

Data Acquisition (DAQ) Subteam Member | University of Virginia

- Implementing SD card logging by reading a shared DAQ struct and writing synchronized data from 17 analog sensor channels to persistent storage.
- Designing sensor-integration hardware and multi-layer PCBs in Altium and KiCad for an all-electric FSAE racecar.

HACKATHON EXPERIENCE

VTHacks 14 **Sep 2026**

2-person team | Embedded Systems & Wireless Sensing

- Produced ESP32 occupancy measurements for a partner-developed Supabase and React/Node.js dashboard visualizing occupancy trends.
- Developed C++ firmware that converted Wi-Fi/Bluetooth packet activity and RSSI into a calibrated 0-100 density score.